

Multiple Choice (10 marks)

1. PreP is an effective strategy for reducing the incidence of HIV in:
 - (a) Women
 - (b) Drug users
 - (c) Pregnant women
 - (d) a. and b. above
2. Laryngeal oedema is usually confined to the larynx above the vocal folds because
 - (a) there are numerous mucous glands in the vestibular folds.
 - (b) the mucosa covering the vocal folds is tightly attached to underlying tissues.
 - (c) fluid will drain rapidly into the thorax below the vocal folds.
 - (d) the mucosa above the vocal folds is more vascular than that below the vocal folds.
3. Deep nucleotide sequencing (NGS) is now cheap enough for general application. What will the technique mainly be useful for?
 - (a) Speedy vaccine development
 - (b) Identifying genetic variation
 - (c) Giving details on virus excretion in symptomless carriers
 - (d) Quickly identifying new viruses
4. All other things being equal, which of the following persons is more likely to show osteoporosis?
 - (a) An older Hispanic American woman
 - (b) An older African American woman
 - (c) An older Asian American woman
 - (d) An older Native American woman
5. Asp 235 Phe in a molecular report indicates that:
 - (a) asparagine has been replaced by phenylalanine.
 - (b) phenylalanine has been replaced by asparagine.
 - (c) aspartic acid has been replaced by phenylalanine.
 - (d) phenylalanine has been replaced by aspartic acid.

True / False (10 marks)

Answer True or False

6. True or False: Proprioceptive nerve endings in synovial joints are located in articular cartilage and the synovial membrane.
7. True or False: A patient with a CVA affecting the left internal capsule will exhibit spastic paralysis in the right leg due to upper motor neuron involvement.
8. True or False: The modern rabies vaccine requires 16 doses administered into the peritoneal cavity for effective immunity.
9. True or False: The risk of abnormality in the child of a mother with untreated phenylketonuria is approximately 10%.
10. True or False: Approximately 10% of adolescent males report having homosexual experiences during their teenage years.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of quasi-experimental research designs in health sciences, particularly in studying differences between two or more groups of people.
12. Discuss the significance of virion RNA acting as mRNA in the replication process of picornaviruses as positive strand RNA viruses. What implications does this have for viral biology?

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